

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course: B. Tech****Branch: Common To All Branches****Semester : I****Subject Code & Name: BTES105 - Energy and Environment Engineering****Max Marks: 60****Date: 25-07-2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is a major disadvantage of steam power stations? A) High efficiency B) Quick start-up time C) High pollution levels D) Low maintenance cost	CO1	1
2	In a nuclear power plant, which element is commonly used as fuel? A) Coal B) Uranium C) Diesel D) Natural Gas	CO1	1
3	Thermoelectric generators work on the principle of: A) Electromagnetic induction B) Thermionic emission C) Seebeck effect D) Nuclear fission	CO1	1
4	Which of the following is NOT a form of renewable energy? A) Wind B) Solar C) Nuclear D) Tidal	CO2	1
5	Ocean Thermal Energy Conversion (OTEC) works based on: A) Salinity differences B) Temperature differences in ocean layers	CO2	1

	C) Wave movement D) Tidal force		
6	The main advantage of fuel cells is: A) High noise during operation B) Emission of greenhouse gases C) Clean and efficient power generation D) Use of radioactive fuel	CO2	1
7	Energy conservation in electric motors can be achieved by: A) Operating them at full load only B) Using variable frequency drives C) Increasing voltage unnecessarily D) Reducing motor size	CO3	1
8	Which of the following is a principle of energy conservation? A) Maximum energy waste B) Maximum energy efficiency C) Ignoring maintenance D) Delaying repairs	CO3	1
9	One of the techniques for lighting energy conservation is: A) Using incandescent lamps B) Painting rooms black C) Installing occupancy sensors D) Reducing natural light	CO4	1
10	Which of the following is a source of particulate emission? A) Electric motors B) Wind turbines C) Diesel engines D) Solar panels	CO4	1
11	Which gas is a major contributor to acid rain? A) Oxygen B) Nitrogen C) Sulphur dioxide D) Helium	CO5	1
12	Noise pollution can be controlled by: A) Increasing engine power B) Using silencers and soundproofing C) Removing vegetation D) Increasing traffic speed	CO5	1

Q. 2	Solve the following.		12
A)	Explain the schematic arrangement and working of a steam power station. Mention its advantages and disadvantages.	CO1	6
B)	List and explain the environmental aspects considered while selecting a site for a nuclear power plant.	CO1	6
Q.3	Solve the following.		12
A)	Describe the working principle of Ocean Thermal Energy Conversion (OTEC) with a neat diagram. Mention its advantages and disadvantages.	CO2	6
B)	Explain the construction and working of a fuel cell. State its advantages over conventional power generation methods.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What are the methods and techniques of energy conservation in compressors, pumps, and fans?	CO3	6
B)	Discuss the principle of energy conservation. Explain the terms "maximum energy efficiency" and "maximum cost effectiveness."	CO3	6
C)	How can energy conservation be achieved in electric furnaces, ovens, and boilers? Give suitable examples.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What are the major sources and effects of air pollution on human health and the environment?	CO4	6
B)	What are air quality standards and how is air pollution measured?	CO4	6
C)	Explain the types of particulate emissions and control measures used to reduce them in industries.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What are the causes and effects of thermal pollution? Suggest suitable control measures.	CO5	6
B)	Write a note on the disposal methods for solid and biomedical wastes.	CO5	6
C)	What are the causes of noise pollution? Explain the effects of noise pollution and measures to control it.	CO5	6
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